Counting Trees

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Overview

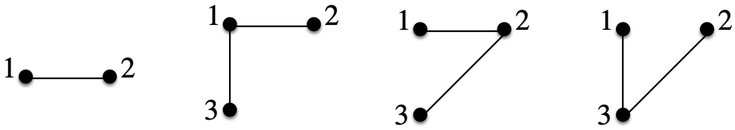
Format

Different format today. We’re going to look at one problem and we’ll see my failed attempts to solve it, along with a solution that actually works.

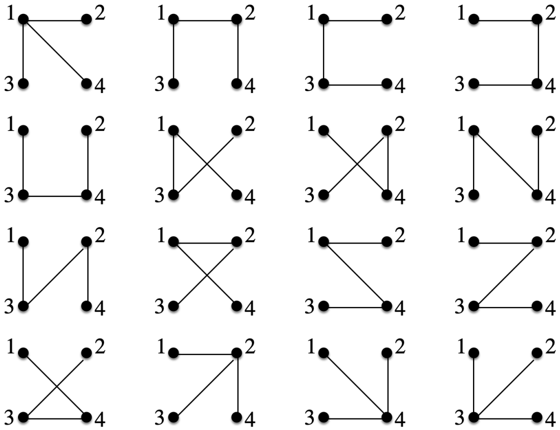
Counting trees

How many ways can we connect *n* vertices together into a tree?

Trees on 2 and trees on 3:



Trees on 4



What we know so far

Trees: Connected, acyclic.

n	trees
2	1
3	3
4	16

Choosing edges

Ok, first thought. A tree on n vertices has $n-1$ edges out of all possible edges:

$$\binom{\binom{n}{2}}{n-1}.$$

n	trees	
2	1	
3	3	
4	20	> 16

We counted cycles that aren't trees.

Ways to add a vertex

So, let's be careful to only generate trees. Here's the thought. Consider a tree on n vertices. We can add vertex $n+1$ and connect it into the tree n different ways. We're guaranteed that the new graph is connected and acyclic (a tree!).

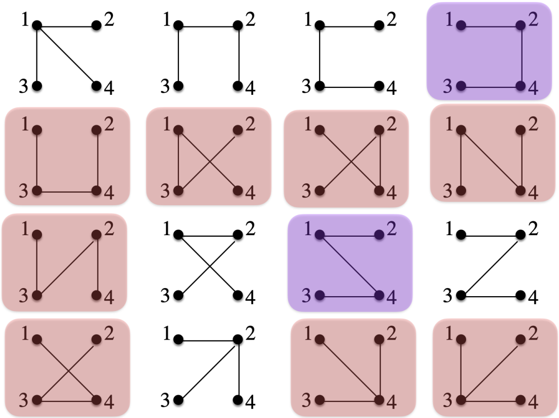
So, 2 vertices make 1 tree. Adding the 3rd vertex creates 2 times more. Adding the 4th vertex creates 3 times more.

Generalizing, we get $(n-1)!$ trees.

n	trees	
2	1	
3	2	< 3
4	6	< 16

Now, we're only counting trees, but we're missing some trees. In particular, we're missing trees where the last vertex is somewhere in the middle.

Generated 4 trees



More careful growing

When we add in vertex $n+1$, the other n vertices might not be a tree. They might be a forest. In general, vertex $n+1$ will have one edge to each connected component in the forest. For the edge to connected component i, it will have a choice of which of the vertices in the connected component to connect to.

You can *almost* write down an expression, but it's complicated and not clear how to simplify it. The basic idea is consider all the ways of making a tree with n' vertices for all $n' \leq n$, then all the ways n vertices can be partitioned into clusters, then sum and multiply...

But even the question of how many partitions there are for n items is hairy. See: https://oeis.org/A000110 .

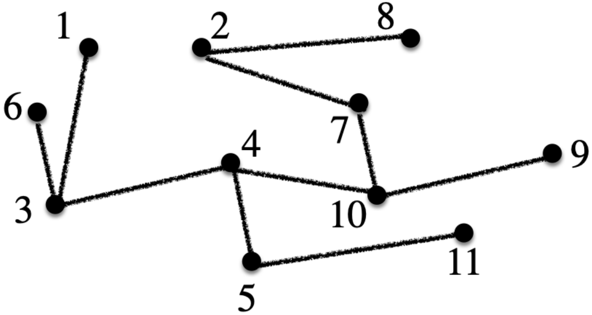
Better way to look at it

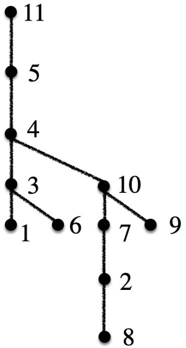
Sometimes there's just a better way to look at it. It's definitely not obvious (to me!). But it's clever and gives a nice clean answer.

1. Create a "normal form" for trees. That way, we can at least notice when two different trees are actually the same.
2. Find a compact encoding for these trees. Here, "compact" means no extraneous information.
3. Then, we can show we have a bijection (!) between trees and the encoding.
4. If we're lucky, it'll be easier to count encodings than trees.

Normal form for trees

Choose vertex *n* to be the root. Order children smallest (left) to biggest (right). There are no other choices.





Encoding a tree

Every vertex has a unique parent. So, we could just give the parent for each vertex: 3 7 4 5 11 3 10 2 10 4.

List $n-1$ numbers (since root has no parent), each with a choice of $n-1$ numbers (can't pick yourself!). That's $(n-1)^{n-1}$.

Anything extraneous? Yes, can encode a loop: 3 1 2 5.

Sanity check

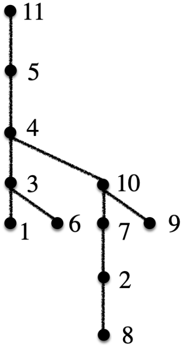
Could it be $(n-1)^{n-1}$?

n	trees	
2	1	
3	4	> 3
4	27	> 16

Yeah, we're definitely overcounting. But we're guaranteed not to undercount. Every tree *has* a representation in this scheme. But some representations do not produce trees. It's not a bijection.

Prufer rule

Repeat the following procedure $n-2$ times. Find the smallest valued leaf. Write down its parent. Delete the leaf.

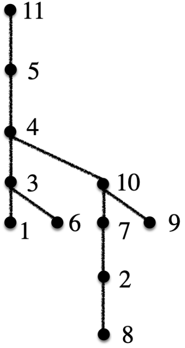


3 3 4 2 7 10 10 4 5

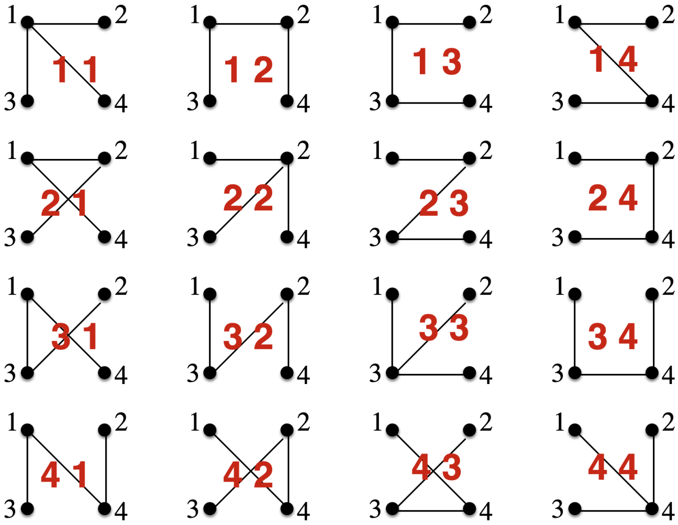
Recover a tree

We can process any tree into a list. But can we recover the tree from the list? Before we prove that we can, let's do an example:

3 3 4 2 7 10 10 4 5

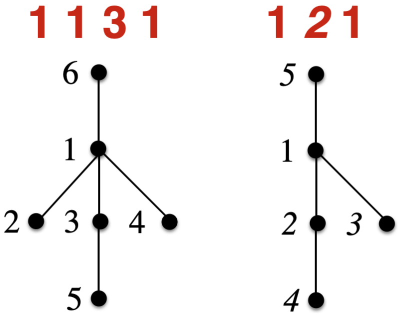


4 by 4



Thinking inductively

Here's a 6-vertex tree in Prufer encoding: 1 1 3 1.



In what sense is it built out of a 5-vertex encoding? Take the vertex x that is the "first" leaf. Here, $x = 2$. Remove it, then renumber the vertices, decrementing anything larger than x. The thing to note is that the resulting tree and encoding still match!

Proof

Theorem: For every string $a \in [1, n]^{n-2}$ ($n \geq 2$), there is a unique tree T.

Proof: Build-down induction on n.

Base Case ($n = 2$): There is only one tree (a 1–2 segment) and only one encoding string (the null string).

Inductive Step ($n + 1$): Consider a string a of length $n - 1$. In the tree T encoded by a, the leaf with the smallest label x must be linked to a_1.

Consider the string a' formed by removing a_1 from a and then subtracting one from every value in a that's larger than x. By the inductive hypothesis, there is a unique tree T_0 constructed from a'. We can construct a unique tree T from T_0 by adding 1 to the values in T_0 that are x or above, then adding the edge (x, a_1).

Summing up

We have a scheme for encoding trees as lists. It always works. We have a scheme for turning lists into trees. It always works. We have a bijection.

How many lists? $n - 2$ choices of numbers from 1 to n. So, n^{n-2}. Does that fit?

n	trees
2	1
3	3
4	16